

september 2026



[noForth website](#)

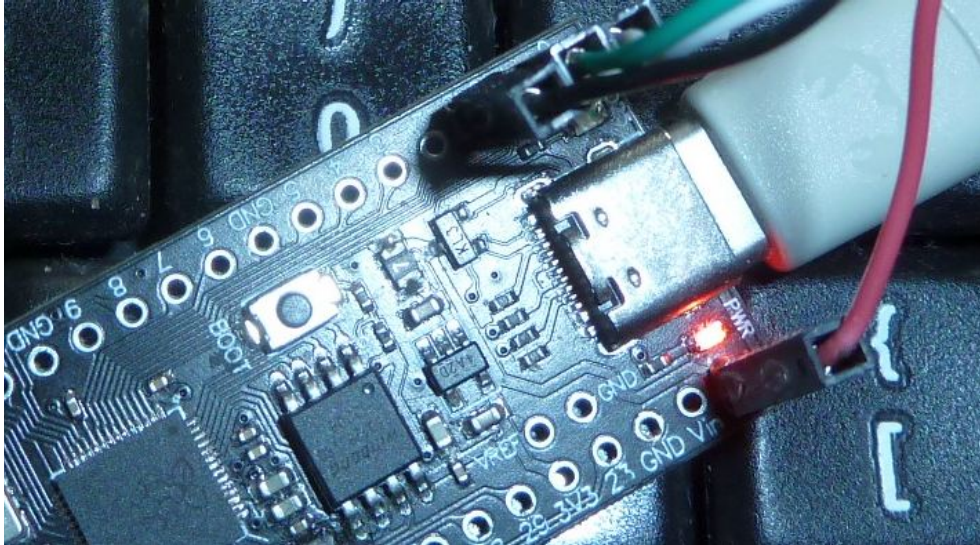
Pico RP2350 dev boards with noForth t(v)(#) (solo/duo)

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In this text we refer to these two documents:

- [RP2350 datasheet](#)
- [Armv8-M Architecture Reference Manual](#)

1. RP2350 Pico dev boards



Core : M33 core with ARMv8-M ISA

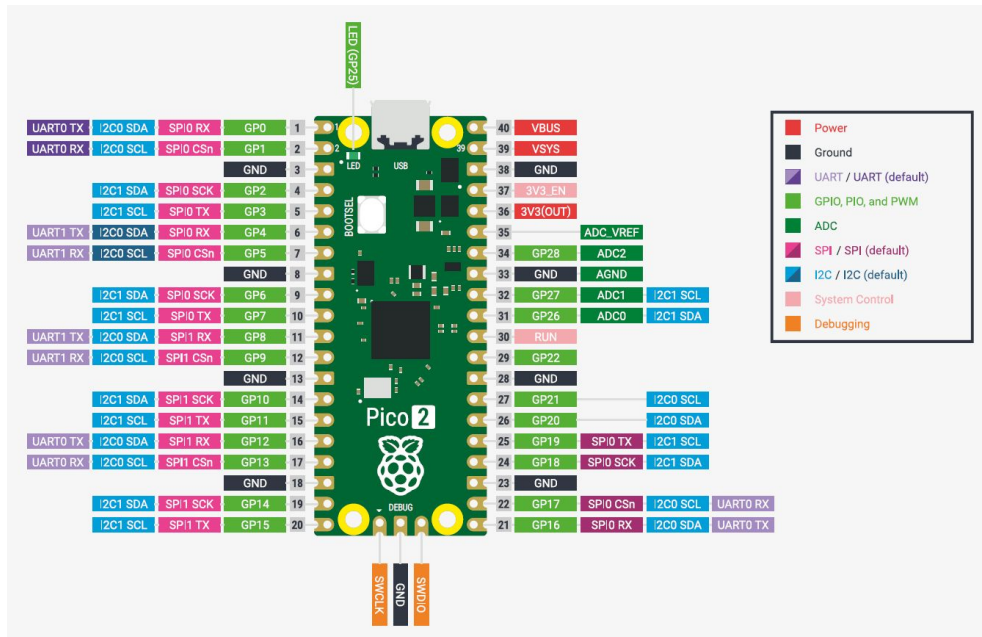
Kit Contents: Various Pico Dev Boards

- Antratek: Pico 2 dev board
- Kiwi: Pico 2 dev board
- Olimex RP2350-PICO2-XXL board
- Pimoroni Pico Plus 2
- Waveshare RP2350-zero
- Waveshare RP2350-Plus
- Waveshare RP2350-PiPico

i/o port connections pico

GPIO

GPIO0	-	UART0 TX	GPIO16	-	...
GPIO1	-	UART0 RX	GPIO17	-	...
GPIO2	-	...	GPIO18	-	...
GPIO3	-	...	GPIO29	-	...
GPIO4	-	UART1 TX	GPIO20	-	...
GPIO5	-	UART1 RX	GPIO21	-	...
GPIO6	-	...	GPIO22	-	...
GPIO7	-	...	GPIO23	-	SMPS
GPIO8	-	...	GPIO24	-	VBUS sense
GPIO9	-	...	GPIO25	-	LED
GPIO10	-	...	GPIO26	-	...
GPIO11	-	...	GPIO27	-	...
GPIO12	-	...	GPIO28	-	...
GPIO13	-	...	GPIO29	-	VSYS
GPIO14	-	...			
GPIO15	-	...			



Pico2 board

Connectors

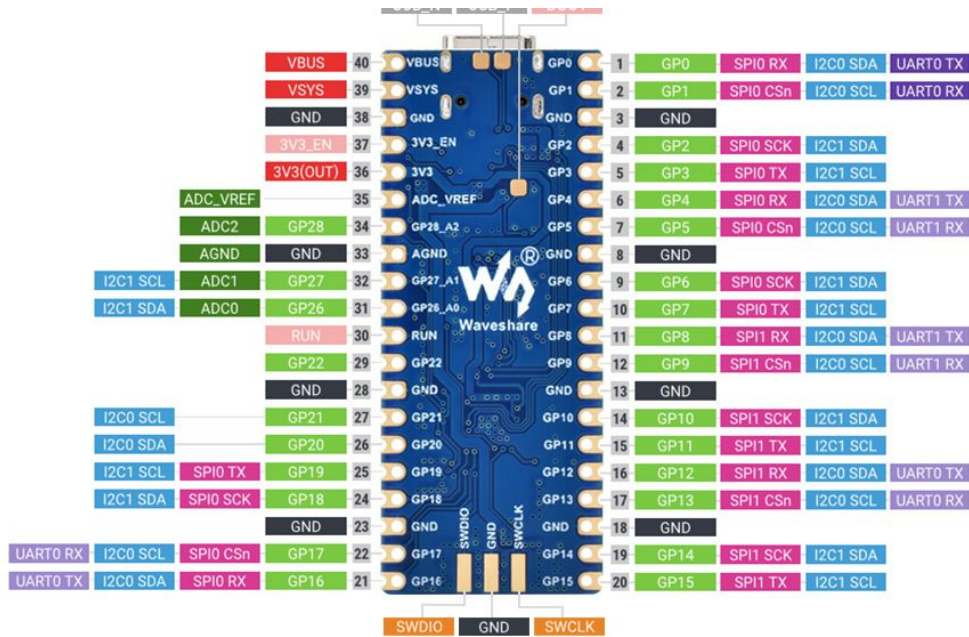
- J0 = i/o GPIO and GND
- J1 = i/o GPIO, RUN, GND, 5V, 3V3, 3V3_EN and ADC_VREF
- J2 = USB-C Programming connector
- J3 = Debug (SWD) port

Hardware Pico

- LED on GPIO25
- Switch BOOTSEL on QSPI_SS
- SPI Flash (16 to 128 Mbit)
- 12MHz crystal

Hardware RP2350-Plus

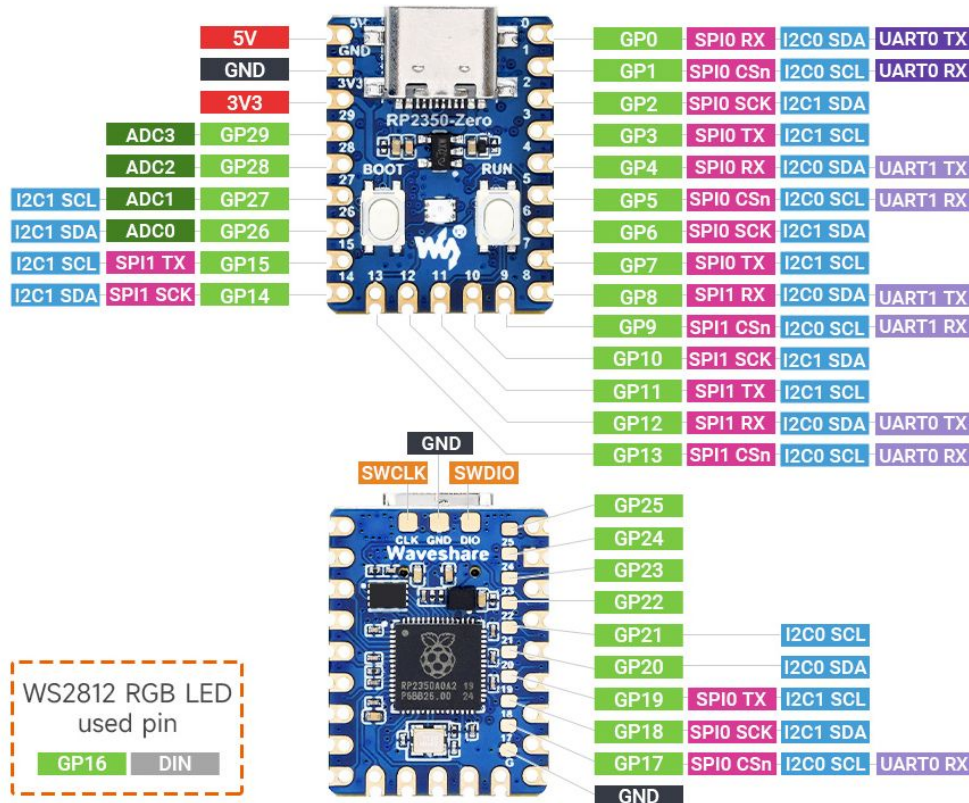
- LED on GPIO25, Lithium battery connector
- Two switches BOOTSEL on QSPI_SS & reset
- SPI Flash (128 Mbit= 16 MByte)
- 12MHz crystal



RP2350-Plus board

Hardware RP2350-zero

- WS2812 LED on GPIO16
- Two switches BOOTSEL on QSPI_SS & reset
- SPI Flash (16 Mbit)
- 12MHz crystal



RP2350-zero board

Hardware RP2350 PiZero

- Lithium battery connector
- Micro SD card connector
- Onboard DVI interface
- PIO-USB port
- Two switches BOOTSEL on QSPI_SS & reset
- SPI Flash (128 Mbit)
- 12MHz crystal



RP2350 PiZero board

2. I/O ports

I/O port addresses

The port registers are memory mapped.

First the base addresses for the I/O-port:

SIO base = D0000000
User bank = 40028000 (Two cells for each pin)

Port register offsets for SIO base

More functionality can be found from page 42ff.

Label	SIO base	Function
CPUID	00	Processor core ID
GPIO_IN	04	Input value for GPIO
GPIO_HI_IN	08	Input value for GPIO & QSPI
GPIO_OUT	10	Output value
GPIO_HI_OUT	14	Output value high
GPIO_OUT_SET	18	Output value set
GPIO_HI_OUT_SET	1C	Output value set high
GPIO_OUT_CLR	20	Output value clear
GPIO_HI_OUT_CLR	24	Output value clear high
GPIO_OUT_XOR	28	Output value toggle
GPIO_HI_OUT_XOR	2C	Output value toggle high
Etc.		

Values for port control IO_BANK0_BASE registers

On page 590 to 593 of the RP2350 datasheet, you find a complete list of all functions that can be chosen for each GPIO pin. The pin control functions starts at offset+4 for GPIO0 and are increased by 8 for each next pin.

GPIO	Function								
	F1	F2	F3	F4	F5	F6	F7	F8	F9
0	SPIO RX	UART0 TX	I2C0 SDA	PWM0 A	SIO	PIO0	PIO1		USB OVCUR DET
1	SPIO CSn	UART0 RX	I2C0 SCL	PWM0 B	SIO	PIO0	PIO1		USB VBUS DET
2	SPIO SCK	UART0 CTS	I2C1 SDA	PWM1 A	SIO	PIO0	PIO1		USB VBUS EN
3	SPIO TX	UART0 RTS	I2C1 SCL	PWM1 B	SIO	PIO0	PIO1		USB OVCUR DET
4	SPIO RX	UART1 TX	I2C0 SDA	PWM2 A	SIO	PIO0	PIO1		USB VBUS DET

- 1: SPI-0 or 1
- 2: UART-0 or 1
- 3: I2C-0 or 1
- 4: PWM-0 to PWM-7
- 5: Standard I/O (Reset state)
- 6: Programmable I/O (PIO-0)
- 7: Programmable I/O (PIO-1)
- 8: Clock in or out
- 9: USB-bus

The reset value for all port control registers is 1F. More info in the 'RP2350 datasheet.PDF' from page 595ff.

3. RS232/USB driver

The USB chip for the SEED board is a dongle with the PL2303TA Prolific USB-chip. This chip needs a specific driver under Windows XP/7/8/10/11. Execute this file: [PL2303-Prolific_DriverInstaller_v1200.exe](#). The default baudrate for the RP2350 controller with noForth is 115200 baud.

There is also a USB CDC driver available that runs on top of [noForth tv#](#). This is a retrofittable option. An off-the-shelf UF2 is also available for simplicity.

4a. noForth t solo (multitask) memory map

RAM 20000000 – 20080000

```
20000000 IVECS      \ 48 vectors
      ... IVECS/    \ Vector table end
200000D0 HOT        \ Uhere starts here
20000200 ORIGIN     \ HERE starts here
\ ... system ...
2000xxxx HERE       \ Current systems end
2003F800 ...        \ Systems end
2003F800 FLYBUF     \ Flyer buffer (400 bytes)
2003FC00 FLYBUF/    \ End of flyer buffer
2003FD00 SP0        \ Data stack (100 bytes grows down)
2003FF00 RP0        \ Return stack (200 bytes grows down)
2003FF00 TIB        \ Input buffer (100 bytes)
20040000 TIB/
20080000 MEMTOP     \ End of RAM memory
```

RAM 20080000 – 20082000

8 kByte of free RAM memory in two 4 kByte banks.

The multitasker version uses a part of this memory for new tasks.

4b. noForth t duo (multitask) memory map

noForth t memory map core-0

RAM 20000000 – 20040000

```
20000000 IVECS      \ 48 vectors
      ... IVECS/    \ Vector table end
200000D0 HOT        \ Uhere starts here
20000200 ORIGIN     \ HERE starts here
\ ... system ...
2000xxxx HERE       \ Current systems end
2003F800 ...        \ Systems end
2003F800 FLYBUF     \ Flyer buffer (400 bytes)
2003FC00 FLYBUF/    \ End of flyer buffer
2003FD00 SP0        \ Data stack (100 bytes grows down)
2003FF00 RP0        \ Return stack (200 bytes grows down)
2003FF00 TIB        \ Input buffer (100 bytes)
20040000 TIB/
20040000 MEMTOP     \ End of core-0 RAM memory
```

noForth t memory map core-1

RAM 20040000 – 20080000

```
20040000 IVECS      \ 48 vectors
... IVECS/         \ Vector table end
200400D0 HOT        \ Uhere starts here
20040200 ORIGIN     \ HERE starts here
\ ... system ...
2004xxxx HERE      \ Current systems end
2007F800 ...        \ Systems end
2007F800 FLYBUF     \ Flyer buffer (400 bytes)
2007FC00 FLYBUF/    \ End of flyer buffer
2007FD00 SP0        \ Data stack (100 bytes grows down)
2007FF00 RP0        \ Return stack (200 bytes grows down)
2007FF00 TIB        \ Input buffer (100 bytes)
20080000 TIB/
20080000 MEMTOP     \ End of core-1 RAM memory
```

RAM 20080000 – 20082000

8 kByte of free RAM memory in two 4 kByte banks.

The multitasker version uses a part of this memory for new tasks.

4c. FLASH ROM (XIP mode) 10000000 – 103FFFFF

```
10000000 Secondary boot code, loads image-1 on startup
10000100 Image-1 \ Cold start image invoked by COLD to
10081000 Image-2 \ Spare image invoked by COLD2
10101000 Free    \ Start of free flash space
```

- TIB/ is a changeable uvalue (Udata).
- Udata is saved by FREEZE too.
- At start-up (and reset or COLD) noForth moves from Flash to RAM.

5. Interrupt vector table

See 'RP2350 datasheet.PDF' for more details (page 82ff) and 'ARMv6-M Architecture Reference Manual.pdf' from page 218ff. The table is pointed to by the VTOR register at PPB_BASE + ED08. Note that the low 8 bits of VTOR are always zero , so it's 256 bytes page aligned.

RP2350 exceptions:

0000	00	- Length of noForth image
0004	01	- Reset vector
0008	02	- NMI handler
000C	03	- Hard fault handler
0010	04	- MemManage handler
0014	05	- Bus fault handler
0018	06	- Usage fault handler
001C	07	- Reserved
0020	08	- Reserved
0024	09	- Reserved
0028	10	- Reserved
002C	11	- Supervisor call handler
0030	12	- Debug Monitor
0034	13	- Reserved
0038	14	- Pend supervisor handler
003C	15	- SysTick handler

RP2350 NVIC IRQ interrupts:

0040	00	- Timer0 IRQ 0	00AC	27	- SIO IRQ FIFO NS
0044	01	- Timer0 IRQ 1	00B0	28	- SIO IRQ BELL NS
0048	02	- Timer0 IRQ 2	00B4	29	- SIO IRQ MTIMECMP
004C	03	- Timer0 IRQ 3	00B8	30	- Clock IRQ
0050	04	- Timer1 IRQ 0	00BC	31	- SPI 0 IRQ
0054	05	- Timer1 IRQ 1	00C0	32	- SPI 1 IRQ
0058	06	- Timer1 IRQ 2	00C4	33	- UART 0 IRQ
005C	07	- Timer1 IRQ 3	00C8	34	- UART 1 IRQ
0060	08	- PWM IRQ wrap 0	00CC	35	- ADC IRQ fifo
0064	09	- PWM IRQ wrap 1	00D0	36	- I2C 0 IRQ
0068	10	- DMA IRQ 0	00D4	37	- I2C 1 IRQ
006C	11	- DMA IRQ 1	00D8	38	- OTP IRQ
0070	12	- DMA IRQ 2	00DC	39	- TRNG IRQ
0074	13	- DMA IRQ 3	00E0	40	- PROC0_IRQ_CTI
0078	14	- USB control IRQ	00E4	41	- PROC1_IRQ_CTI
007C	15	- PIO 0 IRQ 0	00E8	42	- PLL_SYS_IRQ
0080	16	- PIO 0 IRQ 1	00EC	43	- PLL_USB_IRQ
0084	17	- PIO 1 IRQ 0	00F0	44	- POWMAN_IRQ_POW
0088	18	- PIO 1 IRQ 1	00F4	45	- POWMAN_IRQ_TIMER
008C	19	- PIO 2 IRQ 0	00F8	46	- SPAREIRQ_IRQ_0
0090	20	- PIO 2 IRQ 1	00FC	47	- SPAREIRQ_IRQ_1
0094	21	- IO IRQ bank 0	0100	48	- SPAREIRQ_IRQ_2
0098	22	- IO IRQ bank 0 NS	0104	49	- SPAREIRQ_IRQ_3
009C	23	- IO IRQ QSPI	0108	50	- SPAREIRQ_IRQ_4
00A0	24	- IO IRQ QSPI NS	010C	51	- SPAREIRQ_IRQ_5
00A4	25	- SIO IRQ FIFO			
00A8	26	- SIO IRQ BELL			